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Medicinal Macrofungi as Cosmeceuticals: A Review

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ABSTRACT: Macrofungi (mushrooms) have been studied for their nutritional value and medicinal properties. However, progress in the biotechnological application of macrofungi in the cosmetic industry as cosmeceuticals and nutricosmetics remains slow. Currently, the cosmetic industry is in a constant search for valuable natural ingredients or extracts with relevant bioactive properties (e.g., antiaging, anticollagenase, antielastase, antihyaluronidase, antipigmentation, anti-inflammatory, antioxidant, and antityrosinase) to design formulations. Edible medicinal mushrooms are an unlimited source of nutraceuticals and pharmaceuticals. They can be used as a source of ingredients to develop organic cosmeceuticals, nutriceuticals, and nutracosmetics for topical and oral administration.

KEY WORDS: antiaging, bioactive compounds, cosmeceuticals, enzyme inhibitors, medicinal mushrooms and macrofungi, nutracosmetics, nutricosmetics

ABBREVIATIONS: ABA, antibacterial activity; **AbM**, *Agaricus blazei* Murill; **ABTS**, 2,2′-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid); **AD**, atopic dermatitis; **AIA**, anti-inflammatory activity; **AOA**, antioxidant activity; **CWE**, cold-water extract; **DPPH**, 2,2-diphenyl-1-picrylhydrazyl; **EPS**, exopolysaccharide; **FIPs**, fungal immunomodulatory proteins; **FVW**, *Flammulina velutipes* white strain; **FVY**, *F. velutipes* yellow strain; **GF-EPS**, *Grifola frondosa* exopolysaccharides polysaccharide; **GF-MPS**, *G. frondosa* mycelial polysaccharides; **HaCaT**, human keratinocyte cells; **HWCP**, hot-water crude polysaccharide; **HWE**, hot-water extract; **IL**, interleukin; **LPx**, lipid peroxidation; **MRSA**, methicillin-resistant *Staphylococcus aureus*; **PUFA**, polyunsaturated fatty acid; **ROS**, reactive oxygen species; **SIRT1**, Sirtuin 1; **SPF**, sun protection factor; **UV**, ultraviolet

I. INTRODUCTION

The natural process of aging, ultraviolet (UV) radiation, and hormonal alterations affect the skin and other body parts, which results in the generation of reactive oxygen species (ROS) causing oxidative stress.[1] Cosmetic skin and hair care formulations based on naturally derived active ingredients, including plants and fungi, are currently preferred by many consumers for their antioxidant effects.

Currently, the cosmetic industry is in a constant search for bioactive ingredients to design novel cosmeceuticals possessing lower toxicity.[2] Cosmeceuticals are products between cosmetics and pharmaceuticals containing bioingredients with antiaging, anti-inflammatory, antioxidant, and photoprotective effects. They can be extracted from different organisms or obtained biotechnologically (cultivation, fermentation, enzymatic synthesis, etc.).

Recent progress in the biology and biotechnology of macrofungi (commonly referred to as mushrooms), genomics, proteomics, and mycopharmacology has contributed to their use in the medical and food industries.[3] Macrofungi are also considered a valuable source of multifunctional cosmetic ingredients to be exploited as cosmeceuticals.[4–6] They are mainly used in skin and hair care products, and the trend for their industrial use is expected to increase over the years.[7]

Wild or cultivable, edible or inedible, and medicinal or poisonous macrofungi are incorporated in many cosmetic products as producers of bioactive compounds (phenolics, glucans, and other polysaccharides and terpenoids) with antiaging, antibacterial, antifungal, anti-inflammatory, antioxidant, hypocholesterolemic, hypoglycemic, immune-modulating as well as anticollagenase, antielastase, antihyaluronidase,

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antityrosinase, wound-healing, and other activities.[5,8–11] However, there are numerous undescribed or not yet cultivated mushroom species with a potential cosmetic application.[6,12,13]

Edible mushrooms are considered a well-known untapped source of valuable food and nutraceuticals that can improve human health and quality of life. However, inedible or poisonous species (e.g., *Amanita muscaria*, *Ganoderma lucidum*, *Inonotus obliquus*, *Phellinus igniarius*, *Ramaria formosa*, etc.) represent a source of bioactive secondary metabolites to develop mycopharmaceuticals (dietary supplements or nutriceuticals) with health-enhancing and therapeutic properties.[5,9] Macrofungi-derived nutraceuticals and nutriceuticals are obtained from mycelia and/or fruiting bodies and administered *per os* in the form of capsules, tablets, or extracts.[14,15]

Currently, numerous macrofungi-derived commercial products, such as cosmeceuticals (applied topically, creams, lotions, and ointments) and nutracosmetics (administered *per os*), are available in the market. The usage of mushroom products is significantly high due to minimal regulation and safety compared to traditional drugs.[16,17]

The current review reports the main genera and species of macrofungi presently used, or patented to be used, in formulations of cosmeceuticals and nutracosmetics, such as *Agaricus subrufescens*, *A. bisporus*, *Choiromyces maeandriformis*, *Ophiocordyceps sinensis*, *Ganoderma lingzhi*, *G. lucidum*, *Grifola frondosa*, *Hypsizygus ulmarium*, *I. obliquus*, *Lentinula edodes*, *Polyporus* spp., *Phellinus* spp., *Schizophyllum commune*, *Trametes versicolor*, *Tremella fuciformis*, and *Tuber* spp. (Table 1).

We also describe the main ranges of human cosmetic and cosmeceutical products incorporating macrofungi or their extracts in antiaging, antioxidant, hair and skin care, antiacne, or perfume products. Cosmetic brands include Bliss (Hut.com Ltd., Cheshire, UK), La Roche (F. Hoffmann-La Roche Ltd., Basel, Switzerland), Nu-Derm (Obagi Medical Products Inc., Irvine, CA), SensiClear (Mission Scientific Skincare Inc., Gold River, CA), and others (Table 2).

II. MACROFUNGI-DERIVED BIOACTIVE COMPOUNDS AS POTENTIAL COSMECEUTICALS

Over the past years, growing consumer demand for natural low-toxicity organic products with effective skin- and hair-repairing effects have led the cosmetic industry to search for alternative bioingredients. Macrofungi are considered sources of bioactive compounds increasingly used as cosmeceutical ingredients. Scientific data show that medicinal, edible and inedible mushrooms are resources of raw extracts, fractions of extracts, and bioactive molecules (e.g., fatty acids, phenolics, peptides, polysaccharides, terpenoids, vitamins and volatile organic compounds), with some also being enzyme-inhibitory compounds.[3–5,8,10,12] In addition, a large spectrum of bioactive natural sources possess significant antiaging, anti-inflammatory, antioxidant, antipigmentation, antimicrobial, antiwrinkle, moisturizing, and skin-whitening effects, and they may act as inhibitors of collagenase, elastase, and hyaluronidase and may be used to regulate tyrosinase activity (Table 1). Consequently, they may be potentially used as natural cosmeceuticals or nutracosmetics. However, the mechanisms behind the mentioned bioactivities are still unknown.

Antioxidants against harmful free radicals and cellular damage are the most demanded cosmeceuticals since their topical administration is essential for the protection of aging skin. Skin hyperpigmentation is caused by several factors that upregulate melanogenesis, and controlling this unwanted process is a major challenge in dermatology and cosmetology and in developing safe and effective antimelanogenic products. Macrofungi-derived compounds, such as *p*-coumaric acid, can be used to reduce oxidative stress and inflammation and as antiaging and skin-whitening agents.[18] Previous studies have shown that *p*-coumaric acid inhibited melanin synthesis in murine melanoma cells; in human epidermal melanocytes, it showed efficient transdermal delivery and functional efficacy in reducing erythema development and skin UV-induced pigmentation, and therefore can be used as a skin-lightening ingredient.[19]

TABLE 1: Medicinal macrofungi-derived bioactive compounds and their cosmetic effects

Mushroom species	Bioactive compounds	Cosmetic effects	Refs.
Abortiporus biennis (Bull.) Singer	Exopolysaccharides	AOA	22
Agaricus bisporus (J.E. Lange) Imbach	Caffeic, chlorogenic, cinnamic, ferulic, gallic, *p*-hydroxybenzoic, *p*-coumaric and protocatechuic acids, alkaloids, polyphenols	AOA, AIA, ABA, antityrosinase	1,8,38,43,49
A. subrufescens Peck	Polysaccharides, benzenoid aromatic compound	AOA, antipigmentation	29
Amanita muscaria (L.) Lam.	β-D-glucan, FMG	AIA	50
Antrodia cinnamomea T.T. Chang & W.N. Chou.	Benzenoid aromatic compound coenzyme Q_0	ABA, AIA	35
Auricularia polytricha (Mont.)Sacc.	Phenolics, glucans	Antihyaluronidase, antityrosinase, AIA	46
Boletus edulis Bull.	Ergothioneine, glutathione	AOA	51
Calocybe gambosa (Fr.) Donk	Ascorbic acid (vitamin C)	Antiradiation, stimulating collagen synthesis	52
Clitocybe aurantiaca (Wulfen) Studer-Steinhäuslin [= *Hygrophoropsis aurantiaca* (Wulfen) Maire]	Clitocybin A, D	Antielastase, stimulation procollagen synthesis, AOA	53,54
Coprinus comatus (O.F. Muller) Gray	Tocopherols (vitamin E)	Reducing UV damage to skin	52
Ophiocordyceps sinensis (Berkeley) G.H. Sung et al.	Cordycepin	Antipigmentation, anticollagenase, antityrosinase, antielastase, photoprotective	41,42
Cyclocybe aegerita (V. Brig.) Vizzini	Ergothioneine, glutathione	AOA	51
Flammulina velutipes (Curtis) Singer	Glycosides	AIA, AOA	4,36,37
Ganoderma lucidum (Curtis) P. Karst.	Polysaccharides, triterpenes, FIPs	Antityrosinase, melanin inhibitory, AIA	9,25–27,29,38,44
Grifola frondosa (Dicks.) Gray	Extracellular polysaccharides, β-glucan, gallic and tannic acids	AOA, stimulation of collagen biosynthetic, antimelanogenesis	30,49
G. gargal Singer	Ergothioneine	Suppressed TNF-α–mediated activation of NF-κB	55

TABLE 1: (*continued*)

Mushroom species	Bioactive compounds	Cosmetic effects	Refs.
Inonotus sanghuang Sheng H. Wu, T. Hatt. & Y. C. Dai	Chlorogenic acid, icarisid II, isorhamnetin, quercetin, quercitrin, rutin	AOA	56
Laccaria amethystina Cooke	Laccaridiones A, B	Anticollagenase	57
Laetiporus sulphurous (Bull.) Murrill	Phenolic acids	Antihyaluronidase, antityrosinase	45
Lentinula edodes (Berk.) Pegler	Polysaccharides, β-glucan lentinan, eritadenine, phenolics, tocopherols	Antityrosinase, AIA, AOA, ABA	8,21
Macrocybe lobayensis (R. Heim) Pegler & Lodge	Phenolic compounds (cinnamic acid, *p*-coumaric acid, *p*-hydroxybenzoic acid, pyrogallol)	AOA, ABA	40
Macrolepiota procera (Scop.) Singer	Lanostane triterpenoids	Inhibition of NO production	58
Neolentinus lepideus (Fr.) Redhead & Ginns	1,3-Dihydroisobenzofuran-4,5,7-triol,5-methoxy-1,3-dihydroisobenzofuran-4,7-diol, lepidepyrone	Antityrosinase, antihyaluronidase	59
Pleurotus eryngii (DC.) Quél.	Cinnamic, *p*-coumaric and *p*-hydroxy-phenylacetic acids, ergosterol	AOA	60
P. eryngii var. *ferulae* (Lanzi) Saccardo	Ergosterol, nicotinic acid, pleurone, (24*E*)-3β-hydroxycucurbita-5,24-diene-26-oic acid	Antielastase	37,61
P. ostreatus var. *florida*	Phenolics, gallic acid	ABA, AOA, antipigmentation	37
P. ostreatus (Jacq.) P. Kumm.	Alkaloids, phenolics, gallic acid	AMA, AOA, AIA, antipigmentation, antityrosinase	4,8,28,29,37
Phellinus igniarius (L.) Quél.	Nepetidin, betulic acid	NF-κB inhibitory	31
Pholiota nameko (T. Itô) S. Ito & S. Imai	β-D-glucan	AIA	62
Polyporus umbellatus (Pers.) Fr.	Polysaccharides, steroids, ergosteol, terpenoids	AIA, ABA, AOA	23
Ramaria formosa (Pers.) Quél.	Axinysone A, *ent*-aristolone, nambinone A, ramarin A, B	Antielastase	63
Sarcodon leucopus (Pers.) Maas Geest. & Nannf.	Sarcoviolin β, episarcoviolin β	AOA	64

TABLE 1: (*continued*)

Schizophyllum commune Fr.	Phenolics, α- and β-glucans	Antipigmentation, antityrosinase, AOA	32,33
Suillus bellinii (Inzenga) Kuntze	Cinnamic, *p*-hydroxyphenylacetic and protocatechuic acids, ergosterol	AOA	60
Trametes versicolor (L.) Lloyd	Protein SIRT1, phenolic acids	Antityrosinase, antihyaluronidase, AOA	39,45
Tremella fuciformis Berk.	Mannose, polysaccharides, uronic acid	Reduce the water and collagen losses, inhibit the increase of glycosaminoglycans, skin protection, AOA	34
Tuber spp.	Polysaccharides, volatile compounds	AMA, AIA, wound-healing	65
Tylopilus ballouii (Peck) Singer	FMG	Inhibition of superoxide and hydroxyl radicals	66
Volvariella bombycina (Schaeff.) Singer	Isodeoxyhelicobasidin, alkaloids	Antielastase, AMA, AIA, AOA	4

AMA, antimicrobial activity; FMG, fucomannogalactan; NF-κB, nuclear factor-κB; TNF, tumor necrosis factor.

TABLE 2: Medicinal macrofungi incorporated in different cosmetic products

Mushroom species	Cosmetic product	Cosmetic effect
Agaricus subrufescens	Vitamega facial moisturizing mask	Renews and revitalizes skin
Albatrellus confluens (Alb. & Schw. ex Fr.) Kotl.	REN clean skincare evercalm ultra comforting rescue mask	Skin feels smooth with even tone
Ophiocordyceps sinensis	Kosé Sekkisei cream	Moisturizer, suppresses melanin production
Ganoderma lucidum	Eminence organic skin care birch water purifying essence	Antiaging, provides hydration
	Yves Saint Laurent Temps Majeur Elixir De Nuit	Antiaging
	Dr. Andrew Weil for origins mega-mushroom skin-relief micellar cleanser	Antiaging
	Shiseido ultimune power infusing concentrate	Antiaging, anti-inflammatory
	Tela beauty organics encore dri styling cream	Hair protection
Inonotus obliquus (Fr.) Pilát	Supermood the perfect day cream	Antioxidant, skin moisturizer
	Root science, reborn organic face mask	Anti-inflammatory, soothes irritated skin
Lentinula edodes	Griffin+row naturally effective skincare	Anti-inflammatory, skin brightening
	Aveeno positively ageless, daily exfoliating cleanser	Antiaging, skin lifting
	One love organics vitamin D moisture mist	Skin lightener, moisturizer and toner
Pleurotus ostreatus	Hankook Sansim Hyoum Jin firming cream	Enhances elasticity, skin tight and vitalize
Polyporus umbellatus	iS clinical active serum	Brightens and revitalizes skin
Schizophyllum commune	Alqvimia eternal youth maximum recovery face cream	Brightens, unifies skin tone
	Alqvimia eternal youth cream facial máxima regeneración	Antiaging, skin lifting
	Sulwhasoo hydro-aid cream	Hydrates, revitalizes skin and promotes clear
Tremella fuciformis	Eminence organic skin care Birch water purifying essence	Antioxidant, enhances skin elasticity
	Dr. Dennis gross skincare alpha beta pore perfecting cleansing gel	Enhances skin elasticity, skin firming
	La Prairie advanced marine biology night solution	Moisturizes, nourishes, revitalizes
Tuber spp.	Black Orchid Tom Ford, Jo Malone Red Truffle 21, Estée Lauder re-nutriv skin cream, Perigold truffle magnetic face mask	Perfumes with truffle aroma, moisturizes, nourishes

Bioactive peptides with plant and animal origins can have a beneficial effect on multiple physiological pathways in the skin, which makes them ideal candidates for the development of cosmeceuticals. Significant antioxidant and dermatoprotective effects of these peptides after topical application or *per*

os administration have been revealed, which suggests that they can be used in skin care formulations as functional ingredients. However, the mechanisms underlying the skin-protective effect have not been completely elucidated.[20]

Phenolic compounds and tocopherols derived from edible mushrooms *Boletus edulis*, *L. edodes*, and *Xerocomus badius* are the most responsible for their antioxidant activity (AOA).[21] The exopolysaccharide (EPS-1) isolated from fermentation broths of rare, valuable edible medicinal polypore fungus *Abortiporus biennis* also possesses prominent radical-scavenging and macrophage-stimulatory activities *in vitro* and could be explored as a functional ingredient with potential applications in the medical and cosmetic industries.[22] Sclerotia of medicinal polyporoid species *P. umbellatus* have been used in traditional Chinese medicine to treat edema due to its diuretic effect. It contains bioactive compounds responsible for AOA and anti-inflammatory activity (AIA) and has been used as an ingredient in several biotechnology products.[23]

Pharmacological activities of *Ganoderma* proteins, including fungal immunomodulatory proteins (FIPs), as potential cosmeceuticals with anti-aging, anti-allergic, antioxidant, antibacterial, antiviral, immunoregulatory, melanin inhibitory, anti-inflammatory effects have been discussed.[24] Further interdisciplinary studies in metabolomics, proteomics, and system pharmacology will reveal the molecular mechanisms of medicinal effects of *Ganoderma* proteins to assist their usage as cosmetics.[24]

Tyrosinase is an enzyme that has a key role in the formation of melanin. *G. lucidum* extract was found to have potent antityrosinase activity and is part of several cosmetic, skin-lightening beauty products in China.[25,26] *G. lucidum* also contains α/β-D-glucans, alkaloids, triterpenoids (ganoderic acids, ganoderenic acids, ganoderol, ganoderiol, lucidenic acids), sterols (ergosterol), proteins (LZ-8, LZ-9), nucleosides (adenosine, inosine, uridine), and nucleotides (adenine, guanine) with different pharmacological potential.[9] Various marketed products of *G. lucidum* with nutriceutical and cosmeceutical value are already available in the market.[27]

The AOA and antibacterial activity (ABA) of secondary metabolites of *Pleurotus* species have also been reported.[28] The *in vitro* safety of ethanolic extracts derived from *G. lucidum* and *P. ostreatus* incorporated into base cosmetic creams was evaluated. The absence of toxicity for keratinocytes and fibroblasts in a concentration-dependent manner was revealed. Protocatechuic and syringic acids were the only compounds permeating from *G. lucidum* extract. These results have shown the safety of studied extracts as skin care ingredients.[29]

Extracellular polysaccharides derived from medicinal mushroom *Grifola frondosa* (GF-EPS) were applicable in the development of functional cosmeceuticals. The GF-EPS, together with mycelial polysaccharides of *G. frondosa* (GF-MPS), showed AOA, a stimulatory effect on collagen biosynthesis, and antimelanogenesis activities, without significant cytotoxicity. GF-EPS and GF-MPS were shown to be promising cosmetic ingredients.[30] Chemical constituents isolated from medicinal bracket fungus *P. igniarius* could also be used as ingredients to develop anti-inflammatory cosmetic products.[31]

Edible medicinal cultivated mushroom *Sch. commune* contains phenolics and β-glucans beneficial to skin health. However, data on the development of *Sch. commune* as a cosmeceutical ingredient are lacking. The optimal water extraction conditions at 30°C for 1 h were suggested for effective antioxidant and antityrosinase properties of β-glucans in *Sch. commune*.[32] The β-glucans isolated from ethanol extracts of *Sch. commune* also showed potent AOA and may be used as cosmeceuticals.[33]

Edible mushroom *T. fuciformis* has attracted increasing attention due to its various bioactivities. The polysaccharides extracted from *T. fuciformis*, mainly consisting of mannose and uronic acid, could efficiently reduce water and collagen losses of the skin and could inhibit the increase of glycosaminoglycans. The histopathological study showed that oral treatment with polysaccharides alleviated UV-induced structural alterations of the skin, repaired endogenous collagen breakdown, and increased AOA compared to the irradiated control group (without treatment). Thus, polysaccharides from *T. fuciformis* can be used as a skin-protective cosmetic ingridient.[34]

Atopic dermatitis (AD) is an inflammatory skin disease usually accompanied by methicillin-resistant *Staphylococcus aureus* (MRSA) infection due to cutaneous barrier-function damage. Benzenoid aromatic compound coenzyme Q_0 isolated from *Antrodia cinnamomea* possesses ABA and AIA. The strongest anti-MRSA activity was achieved by inhibiting DNA polymerase, topoisomerases, reducing the ribosomal proteins, as well as downregulating the expression of interleukin (IL)-6, chemokine (C-C motif) ligand CCL5, and CCL17 in human keratinocyte cells (HaCaT), respectively. CoQ_0 at 0.5 μg/mL could recover the filaggrin decreased by HaCaT activation to the normal control and is effective in mitigating AD symptoms associated with bacterial infection.[35]

Yellow strain of *Flammulina velutipes* (FVY) has become popular among customers due to its distinct texture different from *F. velutipes* white strain (FVW). The physicochemical properties and AOA of both strains were evaluated. The results showed that FVY-80 possesses the highest AOA based on 2,2-diphenyl-1-picrylhydrazyl (DPPH; IC_{50} = 2.22 mg/mL) and 2,2′-azino-bis-3-ethylbenzthiazoline-6-sulphonic acid (ABTS) radical assay (IC_{50} = 2.04 mg/mL). FVY-80 significantly reduced the ROS content in L929 cells by 55.96% and was suggested as the best source for use as a cosmetic ingredient.[36]

The study of the ABA, AOA, and sun protection factor (SPF) of ethyl acetate extracts of mycelia of selected edible mushrooms revealed ABA against *Staphylococcus epidermidis*, *Escherichia coli*, *Micrococcus luteus*, and *Bacillus megaterium*, as well as high SPF activity (34.02 at 200 μg/mL) and AOA, particularly of cultural broth extracted from *F. velutipes*, *Pleurotus florida*, and *P. ostreatus*. The results showed the presence of bioactive ingredients in these fungi with promising application in cosmeceuticals.[37]

Antimelanogenetic activity and AIA of hot-water extracts (HWEs) of culinary-medicinal mushrooms *Agaricus bisporus* (white and brown), *G. lucidum*, *Hypsizygus marmoreus*, *Pleurotus floridanus*, and *P. pulmonarius* were evaluated. Brown *A. bisporus* reduced intracellular melanin content to the largest extent (up to 57.05 ± 3.90%) without a cytotoxic effect on B16F10 melanoma cells, and it showed antityrosinase activity to 17.93 ± 2.65%. The extract obtained from brown *A. bisporus* had AIA decreasing the levels of NO and tumor necrosis factor-α, and it can be used to develop skin-whitening and anti-inflammatory products. Other mushroom extracts showed moderate activity.[38]

The ethanolic extract from *Agaricus blazei* Murrill (=*A. brasiliensis*) (AbM) was incorporated into a semi-solid base cosmetic cream (safety tests for nutraceutical applications), and cell viability effects of the extract and of the final cream formulation on a keratinocyte cell line (HaCaT) (safety tests for cosmeceutical applications) were studied. The study detected proteins, carbohydrates, and a low-fat content. Overall, 22 fatty acids including 53% of polyunsaturated fatty acids (PUFA) were estimated the most abundant fraction. The cell viabilities of Caco-2 and HT29 cell lines were maintained up to 100 μg/mL. After incorporation into the base cream, a formulation with a pale-yellow color and favorable pH was obtained. The viability of HaCaT cells in the presence of the extract and the final cream formulation was maintained in a concentration-dependent manner, which indicates the safety of this extract for use in the cosmetic industry. The results suggest that AbM residues can be used as an inexpensive and sustainable source of nutraceutical and cosmeceutical ingredients.[25]

Bioactive compounds extracted from macrofungi showed that the extract of *Trametes versicolor* suppressed the UV-induced cellular senescence in human HaCaT cells possibly through augmenting Sirtuin 1 (SIRT1) protein expression.[39] Phenolic compounds (cinnamic acid, *p*-coumaric acid, *p*-hydroxybenzoic acid, pyrogallol) with strong AOA and ABA against *Bacillus subtilis*, *E. coli*, *Listeria monocytogenes*, *Salmonella typhimurium*, and *S. aureus* were detected in *Macrocybe lobayensis*. As a natural antioxidant and antibacterial agent, this fungus was suggested to be used in the pharmaceutical, cosmetic, or food industries.[40]

Macrofungi are also producers of bioactive compounds with antiaging tyrosinase, hyaluronidase, collagenase, and elastase enzyme-inhibitory effects that are incorporated in many cosmetic products (Tables 1 and 2).[5,8,9,11]

Ascomycetous *Cordyceps* (= *Ophiocordyceps*) species is a unique source of bioactive cordycepin with antiaging properties and represents a potential ingredient for cosmeceutical applications.[41]

Evaluation of the skin care benefits of ethyl acetate, ethanol, and hot-water mycelial extracts from valuable medicinal fungus *Ophiocordyceps sinensis* exhibited *in vitro* moderate or significant anticollagenase and photoprotective activity with a SPF up to 25. The ethyl acetate extracts showed the highest antityrosinase and antielastase activity at 0.14–0.47 mg/mL, comparable to arbutin and epigallocatechin gallate. These results revealed the potential of *O. sinensis* as a source of cosmetic bioingredients for skin care products.[42]

The tyrosinase-inhibitory effect, AIA, AOA, and ABA of ethanolic extracts obtained from *A. bisporus*, *P. ostreatus*, and *L. edodes* were reported and their cosmeceutical potential has been evaluated. The phenolic acids and ergosterol were suggested as basic ingredients for cosmetic creams. The findings suggest that the formulations presented in 85–100% of these compounds can further be exploited as cosmeceuticals.[29]

AOA of mushroom-derived biomolecules could provide a skin-lightening effect by inhibition of tyrosinase, while anticollagenase and antielastase activities can help to restore skin elasticity. Three different polysaccharide extracts obtained from *G. lucidum* were screened for their antioxidant, anticollagenase, and antielastase activities. In this study, smaller quantities of proteins, phenols, and flavonoids were detected, as well. None of the analyzed extracts showed overt toxicity to HaCaT cells at concentrations up to 2 mg/mL and displayed superior scavenging ability on ABTS and DPPH radicals. The hot-water crude polysaccharide (HWCP) extract and partially purified fraction were found to be effective inhibitors of lipid peroxidation (LPx) with a two-fold higher inhibition of LPx compared with ascorbic acid (EC_{50} = 1.65 ± 0.08 mg/mL), a common supplement in cosmeceutical formulations. HWCP showed the strongest antityrosinase activity and inhibition of skin extracellular matrix degradation enzymes. Thus, *G. lucidum* may be considered as a promising source of natural cosmeceutical ingredients. The ethanolic extracts obtained from *A. bisporus* and *P. ostreatus* can be used as a source of cosmeceutical ingredients since they possess antioxidant, antityrosinase, and antimicrobial effects. All extracts may be used in free or encapsulated forms in cosmeceutical formulations. The encapsulated forms will provide a controlled release of the cosmetic ingredients, resulting in bioactivity control.[43,44]

Chemical contents, cosmetic properties (e.g., inhibition of tyrosinase and hyaluronidase activity), and SPF values of mycelial extracts derived from *G. applanatum*, *Laetiporus sulphureus*, and *T. versicolor* have been reported.[45] The total amount of phenolic acids in the extracts ranged from 2.69 mg/100 g (*G. applanatum*) to 10.30 mg/100 g (*T. versicolor*) dry weight. The total amount of sterols ranged from 48.4 mg/100 g (*T. versicolor*) to 201.04 mg/100 g (*L. sulphureus*) dry weight. Indoles ranged from 2.9 mg/100 g (*G. applanatum*) to 16.74 mg/100 g (*L. sulphureus*) dry weight. Kojic acid was revealed in the extracts of *L. sulphureus* and *G. applanatum*. *L. sulphureus* extract caused a dose-dependent antihyaluronidase activity, while all other extracts inhibited tyrosinase activity. The extract of *G. applanatum* exhibited an SPF value of approximately 9. The screened species may be used as bioingredients in the formulation of cosmeceuticals.[45]

The assessment of HWE and cold-water extract (CWE) from fruiting bodies of cultivated macrofungi *Auricularia polytricha*, *P. ostreatus*, *G. lucidum*, and *Sch. commune* revealed their huge cosmeceutical potential. Total phenolic, polysaccharide, and glucan contents were determined, and the AOA was estimated. Antihyaluronidase activity was used as an indicator of antiaging and anti-inflammatory properties, while antityrosinase activity was evaluated to assess the whitening property of these extracts. The total polysaccharide content of *P. ostreatus* extracts was the highest, while extracts from *G. lucidum* contained the lowest level of glucan (10.12–10.67%). CWE from *Sch. commune* exhibited 98.15% antityrosinase activity and 94.82% AOA. *G. lucidum* HWE revealed the strongest antihyaluronidase activity (72.78%). The cosmeceutical properties of these mushrooms can be mainly attributed to the combination of polysaccharides and phenolics. Overall, the CWE of *Sch. commune* and the HWE of *G. lucidum* revealed the best outcome and are suggested for further study as potential cosmeceuticals.[46]

Culinary-medicinal mushroom *A. bisporus* is rich in nutrients and minerals, and it possesses potential anticancer, antioxidant, anti-obesity, and anti-inflammatory properties. Its antityrosinase activity and ergothioneine with skin-whitening properties have shown that this fungus is a source of not only nutraceuticals and pharmaceuticals but also cosmeceutical products.[1]

Many commercially available macrofungi-derived cosmetic products have replaced synthetic compounds that have long-term adverse effects.[47] However, the use of macrofungi and their extracts obtained from fruiting bodies or mycelia as cosmeceuticals need to undergo several steps of biotechnological cultivation and production (extraction optimization, assessment of efficacy and safety) to be produced as a cosmetic product. Previous reports regarding stability, compatibility, and toxicological studies of macrofungi also need to be considered.[48]

III. CONCLUSIONS

Macrofungi (mushrooms) have been regarded as a traditional source of natural low- and high-molecular-weight secondary metabolites, and they have been used for centuries to prevent and treat many diseases. However, over the past decade, macrofungi have been increasingly exploited in the search for potential ingredients for use in the cosmetic industry. Several species and their extracts derived from fruiting bodies and/or mycelia are either used or patented to be used as cosmeceuticals and nutracosmetics for their antiaging, anti-wrinkle, skin-whitening and moisturizing, wound-healing, anticollagenase, antielastase, antihyaluronidase, antityrosinase, antihyperpigmentation, anti-inflammatory, and antioxidant effects. In view of the macrofungal biodiversity, only a few taxonomically identified and cultivated species are currently used as cosmeceuticals. Further evaluation of natural resources of macrofungi will promote their exploitation in the cosmetic industry. Advances in fungal biology and biotechnology, genomics, proteomics, metabolomics, systems pharmacology, and molecular cosmetology as well as further study of the molecular mechanisms of medicinal and cosmetic effects of macrofungi will support the discovery of new species for use in the formulation of organic cosmetic products and cosmeceuticals.

Currently, different mushroom-derived cosmeceuticals and nutracosmetics with various properties are available in the multibillion-dollar cosmetic market. Despite the availability of a number of products, the industry is largely unregulated, and consumers should consult a doctor prior to using these products. Recent developments in cosmeceutical biotechnology, including nanosystems and nanodelivery technologies, have been introduced to improve the efficacy and safety of use of organic products in the commercial production of antiaging cosmeceuticals. Further realization of clinical studies, manufacturing of high-quality standardized products, and sustainable production of cosmeceuticals under controlled conditions are warranted.

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REFERENCES

1. Usman M, Murtaza G, Ditta A. Nutritional, medicinal, and cosmetic value of bioactive compounds in button mushroom (Agaricus bisporus): A review. Appl Sci. 2021;11:5943.
2. Kaur IP, Agrawal R. Nanotechnology: A new paradigm in cosmeceuticals. Rec Pat Drug Deliv Formul. 2007;1(2):171–82.
3. Niego AG, Rapior S, Thongklang N, Raspé O, Jaidee W, Lumyong S, Hyde KD. Macrofungi as a nutraceutical source: Promising bioactive compounds and market value. J Fungi. 2021;7:397.
4. Wu Y, Choi MH, Li J, Yang H, Shin HJ. Mushroom cosmetics: The present and future. Cosmetics. 2016;3(3):22.
5. Badalyan SM, Zambonelli A. Biotechnological exploitation of macrofungi for the production of food, pharmaceuticals and

cosmeceuticals. In: Sridhar KR, Deshmukh SK, editors. Advances in macrofungi: Diversity, ecology and biotechnology. Boca Raton, FL: CRC Press; 2019. p. 199–230.
6. Angelini P, Girometta CE, Venanzoni R, Bertuzzi G. Micoterapici per la medicina estetica sistemica: I funghi contro l'invecchiamento cutaneo. Natural. 2021;1:44–53.
7. Pareek A, Jain V, Ratan Y, Sharma S, Jain PK, Dave V. Mushrooming of herbals in new emerging markets of cosmeceuticals. Int J Adv Res Pharm Boil Sci. 2012;2(4):473–80.
8. Taofiq O, González-Paramás AM, Martins A, Barreiro MF, Ferreira ICFR. A mushroom extracts and compounds in cosmetics, cosmeceuticals and nutricosmetics. A review. Ind Crops Prod. 2016;15;90:38–48.
9. Badalyan SM, Gharibyan NG, Kocharyan AE. Perspectives in the usage of bioactive substances of medicinal mushrooms in pharmaceutical and cosmetic industries. Int J Med Mushrooms. 2007;9(3-4):275.
10. Badalyan SM, Barkhudaryan A, Rapior S. Recent progress in research on the pharmacological potential of mushrooms and prospects for their clinical application. In: Agrawal DC, Dhanasekaran M, editors. Medicinal mushrooms: Recent progress in research and development. Singapore: Springer Nature; 2019. p. 1–70.
11. Sharifi-Rad J, Butnariu M, Ezzat SM, Adetunji CO, Imran M, Sobhani SR, Tufail T, Hosseinabadi T, Ramírez-Alarcón K, Martorell M, Maroyi A, Martins N. Mushrooms-rich preparations on wound healing: From nutritional to medicinal attributes. Front Pharmacol. 2020;11:567518.
12. Hyde KD, Bahkali AH, Moslem MA. Fungi—an unusual source for cosmetics. Fungal Divers. 2010;43(1):1–9.
13. Lindequist U. Macrofungi in pharmacy, medicine, cosmetics and nutrition—an appraisal. In: Sridhar KR, Deshmoukh SK, editors. Advances in macrofungi: Pharmaceuticals and cosmeceuticals. Boca Raton, FL: CRC Press; 2021. p. 1–6.
14. Chang ST, Wasser SP. Current and future research trends in agricultural and biomedical applications of medicinal mushrooms and mushroom products (Review). Int J Med Mushrooms. 2018;20(12):1121–33.
15. Wong KH, Cheng PG. Mushroom nutriceuticals in Malaysia. Int J Complem Alt Med. 2018;11(2):75–6.
16. Brandt FS, Cazzaniga A, Hann M. Cosmeceuticals: Current trends and market analysis. Semin Cutan Med Surg. 2011;30(3):141–3.
17. Elkhateeb WA. What medicinal mushroom can do? Chem Res J. 2020;5(1):106–118.
18. Alvarez E, Villa R, Nieto S, Donaire A, García-Verdugo E, Luis SV, Lozano P. The suitability of lipases for the synthesis of bioactive compounds with cosmeceutical applications. Mini Rev Org Chem. 2021;18(4):515–28.
19. Boo YC. p-Coumaric acid as an active ingredient in cosmetics: A review focusing on its anti-melanogenic effects. Antioxidants. 2019;8(8):275.
20. Aguilar-Toalá JE, Hernández-Mendoza A, González-Córdova AF, Vallejo-Cordoba B, Liceaga AM. Potential role of natural bioactive peptides for development of cosmeceutical skin products. Peptides. 2019;122:170170.
21. Heleno SA, Ferreira RC, Antonio AL, Queiroz MJRP, Barros L, Ferreira ICFR. Nutritional value, bioactive compounds and antioxidant properties of three edible mushrooms from Poland. Food Biosci. 2015;11:48–55.
22. Long Z, Liu H, Li J, Sun J, Xue Y, Hu Z, Su Z, Xu C, Yan JK. Preliminary characterization of exopolysaccharides produced by Abortiporus biennis in submerged fermentation. Sains Malays. 2019;48(12):2633–40.
23. Bandara AR, Rapior S, Bhat DJ, Kakumyan P, Chamyuang S, Xu J, Hyde KD. Polyporus umbellatus, an edible-medicinal cultivated mushroom with multiple developed health-care products as food, medicine and cosmetics: A review. Cryptogam Mycol. 2015;36(1):3–42.
24. Li LD, Mao PW, Shao KD, Bai XH, Zhou XW. Ganoderma proteins and their potential applications in cosmetics. Appl Microbiol Biotechnol. 2019;103(23-24):9239–50.
25. Jiang L. Ganoderma lucidum (reishi mushroom): Potential application as health supplement and cosmeceutical ingredient. Global J Res Anal. 2015;4:124–5.
26. Chien CC, Tsai ML, Chen CC, Chang SJ, Tseng CH. Effects of tyrosinase activity by the extracts of Ganoderma lucidum and related mushrooms. Mycopathologia. 2008;166:117–20.
27. Ahmad R, Riaz M, Khan A, Aljamea A, Algheryafi M, Sewaket D, Alqathama A. Ganoderma lucidum (reishi) an edible mushroom; a comprehensive and critical review of its nutritional, cosmeceutical, mycochemical, pharmacological, clinical, and toxicological properties. Phytother Res. 2021;35(11):6030–62.
28. Cardoso RVC, Carocho M, Fernandes Â, Pinela J, Stojković D, Soković M, Zied DC, Cobos JDV, González-Paramás AM, Ferreira ICFR, Barros L. Antioxidant and antimicrobial influence on oyster mushrooms (Pleurotus ostreatus) from substrate supplementation of calcium silicate. Sustainability. 2021;13:5019.
29. Taofiq O, Heleno SA, Calhelha RC, Alves MJ, Barros L, Barreiro MF, González-Paramás AM, Ferreira IC. Development of mushroom-based cosmeceutical formulations with anti-inflammatory, anti-tyrosinase, antioxidant, and antibacterial properties. Molecules. 2016;21(10):1372.
30. Kim SW, Hwang HJ, Lee BC, Yun JW. Grifola frondosa polysaccharides as cosmeceuticals. Food Technol Biotechnol. 2007;45(3):295–305.

31. Jiang Z, Jin M, Zhou W, Li R, Zhao Y, Jin X, Li G. Anti-inflammatory activity of chemical constituents isolated from the willow bracket medicinal mushroom Phellinus igniarius (Agaricomycetes). Int J Med Mushrooms. 2018;20(2):119–28.
32. Abd Razak DL, Fadzil NH, Jamaluddin A, Abd Rashid NY, Sani NA, Manan MA. Effects of different extracting conditions on anti-tyrosinase and antioxidant activities of Schizophyllum commune fruit bodies. Biocatal Agric Biotechnol. 2019;19:101116.
33. Phuket SR, Sangkaew T, Chanapan P, Techaoei S. Biological activity of ß-glucans from edible mushroom, Schizophyllum commune in Thailand. Int J Appl Pharm. 2019;11(5):110–2.
34. Wen L, Gao Q, Ma CW, Ge Y, You L, Liu RH, Fu X, Liu D. Effect of polysaccharides from Tremella fuciformis on UV-induced photoaging. J Funct Foods. 2016;20:400–10.
35. Chou WL, Lee TH, Huang TH, Wang PW, Chen YP, Chen CC, Chang ZY, Fang JY, Yang SC. Coenzyme Q0 from Antrodia cinnamomea exhibits drug-resistant bacteria eradication and keratinocyte inflammation mitigation to ameliorate infected atopic dermatitis in mouse. Front Pharmacol. 2019;10:1445.
36. Hu YN, Sung TJ, Chou CH, Liu KL, Hsieh LP, Hsieh CW. Characterization and antioxidant activities of yellow strain Flammulina velutipes (Jinhua mushroom) polysaccharides and their effects on ROS content in L929 cell. Antioxidants. 2019;8(8):298.
37. Saraswat A, Mathur P, Sanyal D. Assessing cosmeceuticals properties of some macrofungi for improved skin care. Int J Pharm Sci. 2020;12(2):15–19.
38. Saad HM, Sim KS, Tan YS. Antimelanogenesis and anti-inflammatory activity of selected culinary-medicinal mushrooms. Int J Med Mushrooms. 2018;20(2):141–53.
39. Chong Z, Matsuo H, Kuroda M, Yamashita S, Parajuli GP, Manandhar HK, Shimizu K, Katakura Y. Mushroom extract inhibits ultraviolet B-induced cellular senescence in human keratinocytes. Cytotechnology. 2018;70(3):1001–8.
40. Khatua S, Acharya K. Functional ingredients and medicinal prospects of ethanol extract from Macrocybe lobayensis. Pharmacogn J. 2018;10(6):1154–1158.
41. Kunhorm P, Chaicharoenaudomrung N, Noisa P. Enrichment of cordycepin for cosmeceutical applications: Culture systems and strategies. Appl Microbiol Biotechnol. 2019;103(4):1681–91.
42. Cheng WY, Wei XQ, Siu KC, Song AX, Wu JY. Cosmetic and skincare benefits of cultivated mycelia from the Chinese caterpillar mushroom, Ophiocordyceps sinensis (Ascomycetes). Int J Med Mushrooms. 2018;20(7):623–36.
43. Taofiq O, Heleno SA, Calhelha RC, Fernandes IP, Alves MJ, Barros L, González-Paramás AM, Ferreira ICFR, Barreiro MF. Mushroom-based cosmeceutical ingredients: Microencapsulation and in vitro release profile. Ind Crops Prod. 2018;124:44–52.
44. Kozarski M, Klaus A, Jakovljević D, Todorović N, Wan-Mohtar WAAQI, Nikšić M. Ganoderma lucidum as a cosmeceutical: Antiradical potential and inhibitory effect on hyperpigmentation and skin extracellular matrix degradation enzymes. Arch Biol Sci. 2019;71(2):253–64.
45. Sułkowska-Ziaja K, Grabowska K, Apola A, Kryczyk-Poprawa A, Muszyńska B. Mycelial culture extracts of selected wood-decay mushrooms as a source of skin-protecting factors Biotechnol Lett. 2021;43:1051–61.
46. Abd Razak DL, Jamaluddin A, Abd Rashid NY, Sani NA, Manan MA. Assessment of cosmeceutical potentials of selected mushroom fruit body extracts through evaluation of antioxidant, anti-hyaluronidase and anti-tyrosinase activity. J Multidiscipl Sci. 2020;3:329–42.
47. Sujarit K, Suwannarach N, Kumla J, Lomthong T. Mushrooms: Splendid gifts for the cosmetic industry. Chiang Mai J Sci. 2021;48(3):699–725.
48. Taofiq O, Barreiro MF, Ferreira ICFR. The role of bioactive compounds and other metabolites from mushrooms against skin disorders—a systematic review assessing their cosmeceutical and nutricosmetic outcomes. Curr Med Chem. 2020;27(41):6926–65.
49. Wu JY, Siu KC, Geng P. Bioactive ingredients and medicinal values of Grifola frondosa (maitake). Foods. 2021;10:95.
50. Ruthes AC, Carbonero ER, Machado Córdova M, Baggio CH, Lanzi Sassaki G, Gorin PAJ, Soares Santos AR, Iacomini M. Fucomannogalactan and glucan from mushroom Amanita muscaria: Structure and inflammatory pain inhibition. Carbohydr Polym. 2013;98(1):761–9.
51. Kalaras MD, Richie JP, Calcagnotto A, Beelman RB. Mushrooms: A rich source of the antioxidants ergothioneine and glutathione. Food Chem. 2017;233:429–33.
52. Vaz JA, Barros L, Martins A, Santos-Buelga C, Vasconcelos MH, Ferreira ICFR. Chemical composition of wild edible mushrooms and antioxidant properties of their water soluble polysaccharidic and ethanolic fractions. Food Chem. 2011;126(2):610–16.
53. Kim YH, Ryoo IJ, Choo SJ, Xu GH, Lee S, Seok SJ, Bae K, Yoo ID. Clitocybin D, a novel human neutrophil elastase inhibitor from the culture broth of Clitocybe aurantiaca. J Microbiol Biotechnol. 2009;19(10):1139–41.
54. Lee JE, Lee IS, Kim KC, Yoo ID, Yang HM. ROS scavenging and anti-wrinkle effects of clitocybin A isolated from the mycelium of the mushroom Clitocybe aurantiaca. J Microbiol Biotechnol. 2017;27:933–8.
55. Ito T, Kato M, Tsuchida H, Harada E, Niwa T, Osawa T. Ergothioneine as an anti-oxidative/anti-inflammatory component in several edible mushrooms. Food Sci Technol Res. 2011;17(2):103–10.

56. Liu K, Xiao X, Wang J, Chen CYO, Hu H. Polyphenolic composition and antioxidant, antiproliferative, and antimicrobial activities of mushroom Inonotus sanghuang. LWT Food Sci Technol. 2017;82:154–61.
57. Berg A, Reiber K, Dörfelt H, Walther G, Schlegel B, Gräfe U. Laccaridiones A and B, new protease inhibitors from Laccaria amethystea. J Antibiot Res. 2000;53(11):1313–6.
58. Chen HP, Zhao ZZ, Li ZH, Huang Y, Zhang SB, Tang Y, Yao JN, Chen L, Isaka M, Feng T, Liu JK. Anti-proliferative and anti-inflammatory lanostane triterpenoids from the Polish edible mushroom Macrolepiota procera. J Agric Food Chem. 2018;66:12:3146–54.
59. Ishihara A, Ide Y, Bito T, Ube N, Endo N, Sotome K, Maekawa N, Ueno K, Nakagiri A. Novel tyrosinase inhibitors from liquid culture of Neolentinus lepideus. Biosci Biotechnol Biochem. 2018;82(1):22–30.
60. Souilem F, Fernandes Â, Calhelha RC, Barreira JCM, Barros L, Skhiri F, Martins A, Ferreira ICRF. Wild mushrooms and their mycelia as sources of bioactive compounds: Antioxidant, anti-inflammatory and cytotoxic properties. Food Chem. 2017;230:40–8.
61. Lee IS, Ryoo IJ, Kwon KY, Ahn JS, Yoo ID. Pleurone, a novel human neutrophil elastase inhibitor from the fruiting bodies of the mushroom Pleurotus eryngii var. ferulae. J Antibiot. 2011;64(8):587–9.
62. Abreu H, Fogagnoli Simas F, Ribeiro Smiderle F, Sovrani V, Dallazen JL, Maria-Ferreira D, Werner MF, Cordeiro LMC, Iacomini M. Gelling functional property, anti-inflammatory and antinociceptive bioactivities of β-D-glucan from the edible mushroom Pholiota nameko. Int J Biol Macromol. 2019;122:1128–35.
63. Kim KC, Lee IS, Yoo ID, Ha BJ, Sesquiterpenes from the fruiting bodies of Ramaria formosa and their human neutrophil elastase inhibitory activity. Chem Pharm Bull. 2015;63(7):554–7.
64. Ma K, Han J, Bao L, Wei T, Liu H. Two sarcoviolins with antioxidative and α-glucosidase inhibitory activity from the edible mushroom Sarcodon leucopus collected in Tibet. J Nat Prod. 2014;77(4):942–7.
65. Thomas P, Elkhateeb W, Daba G. Industrial applications of truffles and truffle-like fungi. In: Sridhar KR, Deshmukh SK, editors. Advances in macrofungi: Industrial avenues and prospects. Boca Raton, FL: CRC Press; 2021. p. 82–8.
66. Lima ATM, Santos MN, de Souza LAR, Pinheiro TS, Paiva AAO, Dore CMPG, Costa MSSP, Santos ND, Baseia YG, Araújo RM, Leite EL. Chemical characteristics of a heteropolysaccharide from Tylopilus ballouii mushroom and its antioxidant and anti-inflammatory activities. Carbohydr Polym. 2016;144:400–9.